



IPv6 Addresses

IPv6 is the successor of IPv4. In contrast to IPv4, the **IPv6** address is **128** bit long. The **prefix** identifies the host and network parts. The Internet Assigned Numbers Authority (**IANA**) is responsible for assigning IPv4 and IPv6 addresses and their associated network portions. In the long term, **IPv6** is expected to completely replace IPv4, which is still predominantly used on the Internet. In principle, however, IPv4 and IPv6 can be made available simultaneously (**Dual Stack**).

IPv6 consistently follows the **end-to-end** principle and provides publicly accessible IP addresses for any end devices without the need for NAT. Consequently, an interface can have multiple IPv6 addresses, and there are special IPv6 addresses to which multiple interfaces are assigned.

IPv6 is a protocol with many new features, which also has many other advantages over IPv4:

- Larger address space
- Address self-configuration (SLAAC)
- Multiple IPv6 addresses per interface

- Faster routing
- End-to-end encryption (IPsec)
- Data packages up to 4 GByte

Features	IPv4	IPv6
Bit length	32-bit	128 bit
OSI layer	Network Layer	Network Layer
Adressing range	~ 4.3 billion	~ 340 undecillion
Representation	Decimal	Hexadecimal
Prefix notation	10.10.10.0/24	fe80::dd80:b1a9:6687:2d3b/64
Dynamic addressing	DHCP	SLAAC / DHCPv6
IPsec	Optional	Mandatory

There are three different types of IPv6 addresses:

Type	Description
Unicast	Addresses for a single interface.

Type	Description
Anycast	Addresses for multiple interfaces, where only one of them receives the packet.
Multicast	Addresses for multiple interfaces, where all receive the same packet.

Note: Contrary to IPv4, IPv6 eliminates the broadcast address. Instead, IPv6 uses multicast addresses to support discovery and communication with multiple nodes.

Hexadecimal System

The **hexadecimal system** (**hex**) is used to make the binary representation more readable and understandable. We can only show **10** (**0-9**) states with the decimal system and **2** (**0** / **1**) with the binary system by using a single character. In contrast to the binary and decimal system, we can use the hexadecimal system to show **16** (**0-F**) states with a single character.

Decimal	Hex	Binary
1	1	0001
2	2	0010
3	3	0011

Decimal	Hex	Binary
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Let's look at an example with an IPv4, at how the IPv4 address (`192.168.12.160`) would look in hexadecimal representation.

Representation	1st Octet	2nd Octet	3rd Octet	4th Octet
Binary	1100 0000	1010 1000	0000 1100	1010 0000
Hex	C0	A8	0C	A0
Decimal	192	168	12	160

In total, the IPv6 address consists of `16 bytes` . Because of its length, an `IPv6` address is represented in a `hexadecimal` notation. Therefore the `128 bits` are divided into `8 blocks` multiplied by 16 bits (or `4 hex` numbers). All four hex numbers are grouped and separated by a colon (`:`) instead of a simple dot (`.`) as in IPv4. To simplify the notation, we leave out leading at least `4` zeros in the blocks, and we can replace them with two colons (`::`).

An IPv6 address can look like this:

- Full IPv6: `fe80:0000:0000:0000:dd80:b1a9:6687:2d3b/64`
- Short IPv6: `fe80::dd80:b1a9:6687:2d3b/64`

An IPv6 address consists of two parts:

- `Network Prefix` (network part)

- **Interface Identifier** also called **Suffix** (host part)

The **Network Prefix** identifies the network, subnet, or address range. The **Interface Identifier** is formed from the **48-bit MAC** address (which we will discuss later) of the interface and is converted to a **64-bit address** in the process. The default prefix length is **/64**. However, other typical prefixes are **/32**, **/48**, and **/56**. If we want to use our networks, we get a shorter prefix (e.g. **/56**) than **/64** from our provider.

In RFC 5952, the aforementioned IPv6 address notation was defined:

- All alphabetical characters are always written in lower case.
- All leading zeros of a block are always omitted.
- One or more consecutive blocks of **4 zeros** (hex) are shortened by two colons (**::**).
- The shortening to two colons (**::**) may only be performed **once** starting from the left.



12 / 21 Sections



Mark Complete & Next